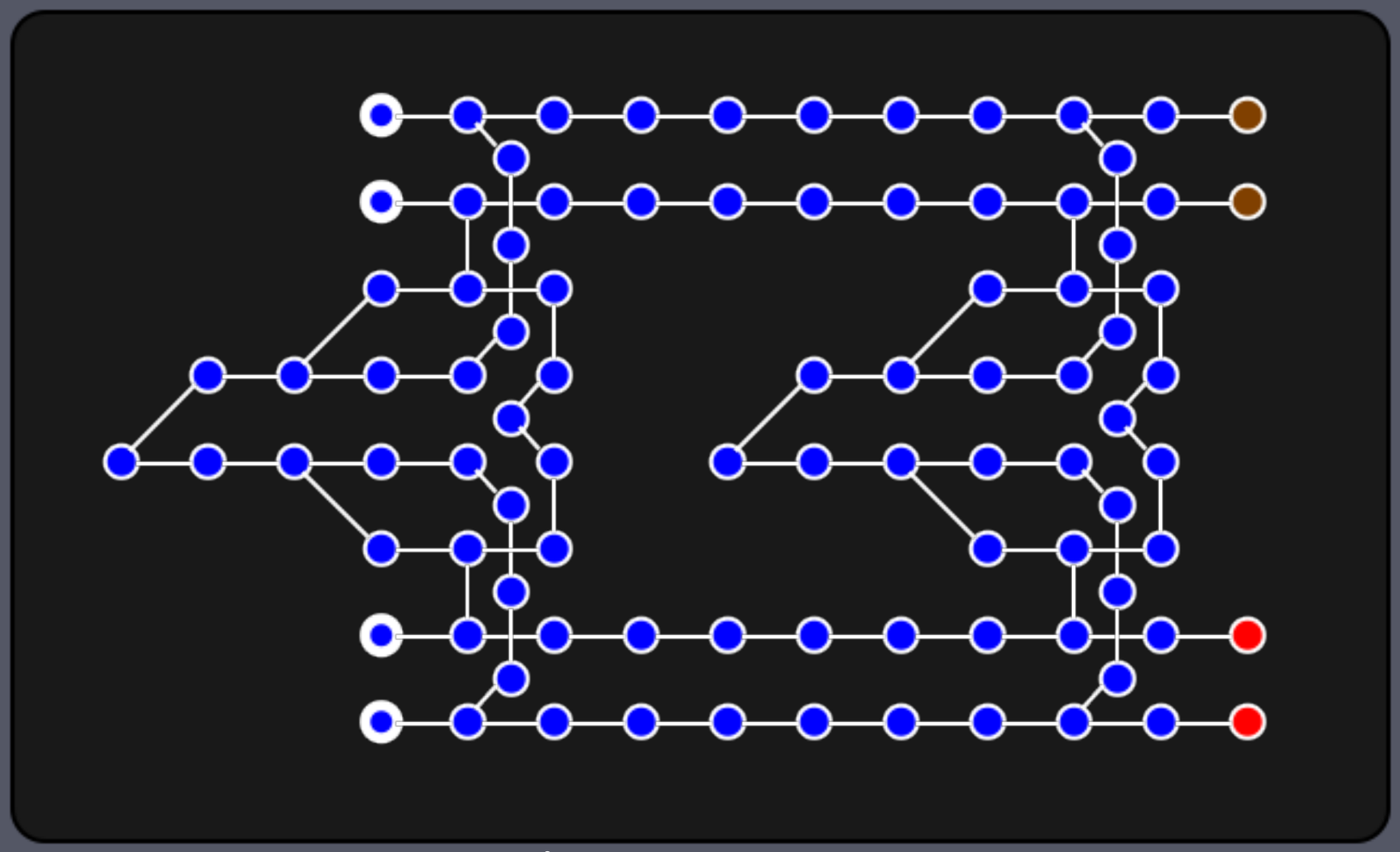
Quantum Error Correcting Codes For Locavores



Dave Bacon* and Jonathan Shi
University of Washington

*= decohered to Google

Thanks Daniel!

VOLUME 82, NUMBER 22 PHYSICAL REVIEW LETTERS 31 MAY 1999

Concatenating Decoherence-Free Subspaces with Quantum Error Correcting Codes

D. A. Lidar,[1] D. Bacon,[1,2] and K. B. Whaley[1]
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[2]*Physics Department, The University of California, Berkeley, California 94720*
(Received 28 September 1998)

An operator sum representation is derived for a decoherence-free subspace (DFS) and used to (i) show that DFS's are the class of quantum error correcting codes (QECC's) with fixed, *unitary* recovery operators and (ii) find explicit representations for the Kraus operators of collective decoherence. We demonstrate how this can be used to construct a concatenated DFS-QECC code which protects against collective decoherence perturbed by independent decoherence. The code yields an error threshold which depends only on the perturbing independent decoherence rate. [S0031-9007(99)09301-1]

PACS numbers: 03.67.Lx, 03.65.Bz, 03.65.Fd, 89.70.+c



But Why?



20 . "Concatena… Rev. Lett. **82**, 4556 (1999…

19 . "Calculatin… ", Phys. Rev. E 59, 2…

18 . "Patter… Rev. E **59**, R4713 (199…

1… …S. Ma…

16 . "Decoherence-… …998), b… D.A. Lidar, I.L. Chuang, …

15 . "Fractality in… …lcai, … Lidar, and D. Avr…

14 . "Is Nature … D.A. Lidar, and D. Avnir [pdf…

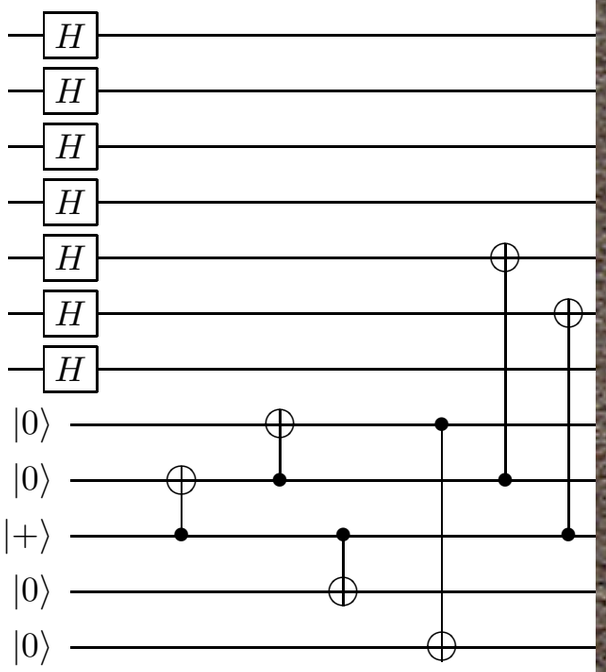
13 . "Is the Geor… A. Lidar, and O. Malcai [pdf]

12 . "Inversion of Randomi… , Inverse Problems **14**, 1299 (1998) [pdf]

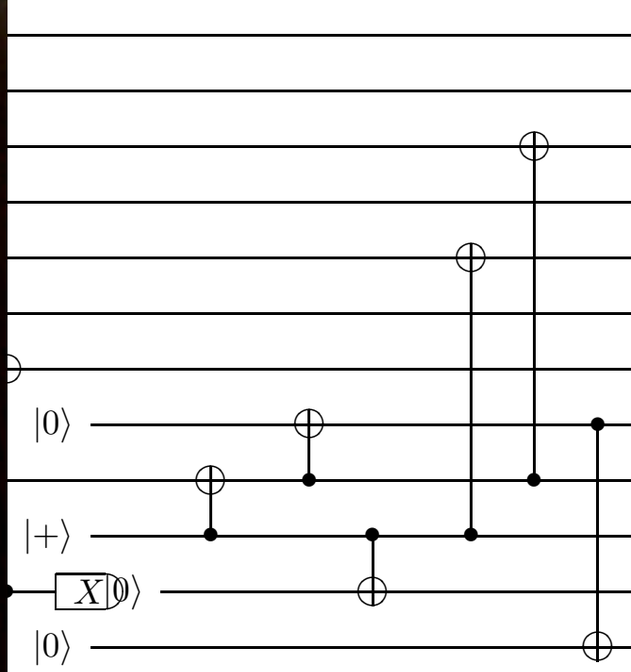
11 . "Atom Scattering from Disordered Surfaces in the Sudden Approximation: Double Collisions Effects and Quantum Liquids", Surf. Sci. **411**, 231 (1998), by D.A. Lidar (Hamburger) [pdf]

10 . "Structure Determination of Disordered Metallic Sub-Monolayers by Helium Scattering: A Theoretical and Experimental Study", Surf. Sci. **410**, L721 (1998), by A.T. Yinnon, D.A. Lidar (Hamburger), R.B. Gerber, P. Zeppenfeld, M. Krzyzowski, and G. Comsa [pdf]

Quar ection





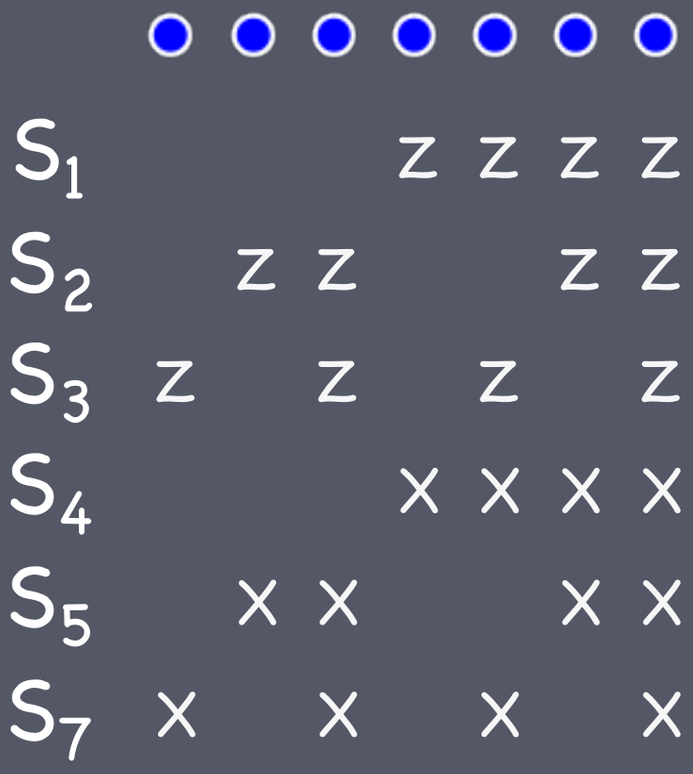


Compl be for
pr on

I CANZ FIT N XPERIMENTALST'S BOX?

Stabilizer Codes

[[7,1,3]] Steane code

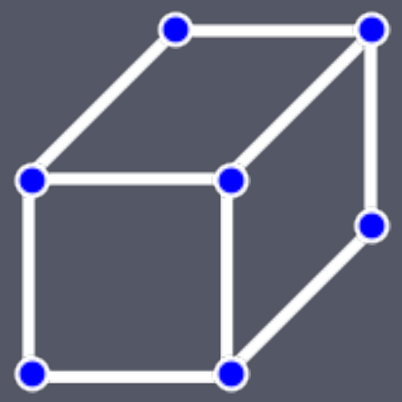


row = stabilizer

column = qubit

QEC Recipe

1. Measure 6 stabilizer operators
2. Conditional on measurement diagnose (and possibly fix) single qubit error



[[# qubits, # encoded qubits, distance]]

Cor ay

- ...

z z z

x x x

z z

z z

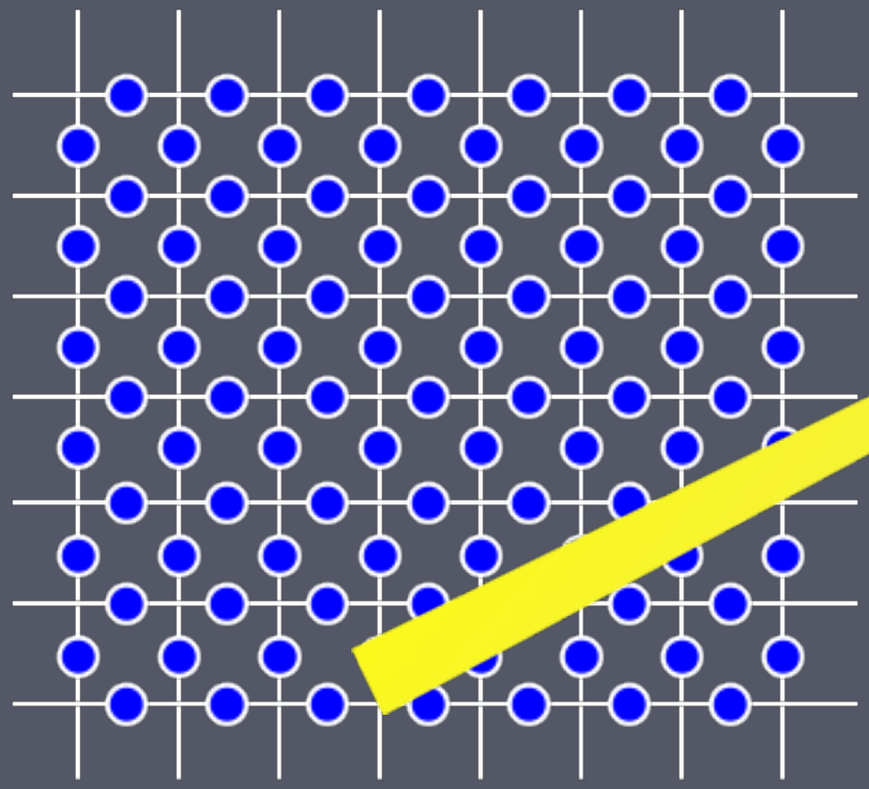
x x

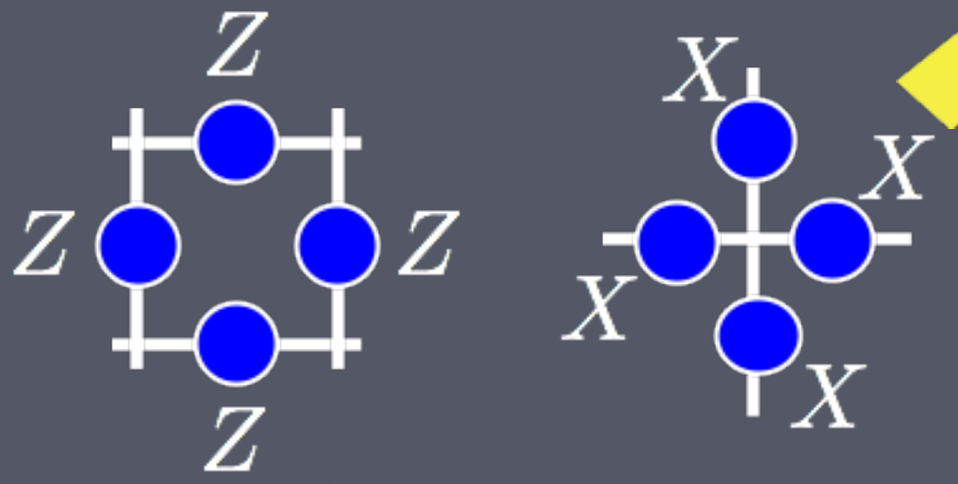
x x



Conc
dista

Rays of Hope





plaquette vertex

Surface code QEC

1. Measure plaquette and vertex operators

2. Classically post-process measurement results to diagnose error

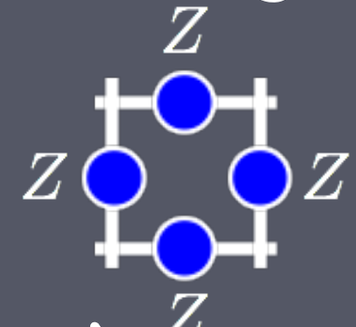
s-Local and s-Neighboring

weight of Pauli operator = # of operators ≠ I

➡ $I^{\otimes}X^{\otimes}X^{\otimes}Y^{\otimes}Y^{\otimes}I^{\otimes}Z^{\otimes}I$ has weight 5

s-Local code: exists a set of stabilizer generators whose weight ≤s

➡ surface code is 4-local



s-Neighboring code (in D dimensions): exists a set of stabilizer generators where all ≠ I operators are within a distance s of each other on some lattice of qubits in D spatial dimensions

➡ surface code is 2-neighboring in 2D

Lament



Why can't all codes be s-neighboring in two or three spatial dimensions?

Main Result

For every [[n,k,d]] stabilizer code, there exists a

[[N,k,r,d]] sta , $N=O(n^2)$,

whose syndr n be made

using s-neigl (s constant) in

>1 spatial dir qubits)

Every st t



Main Result

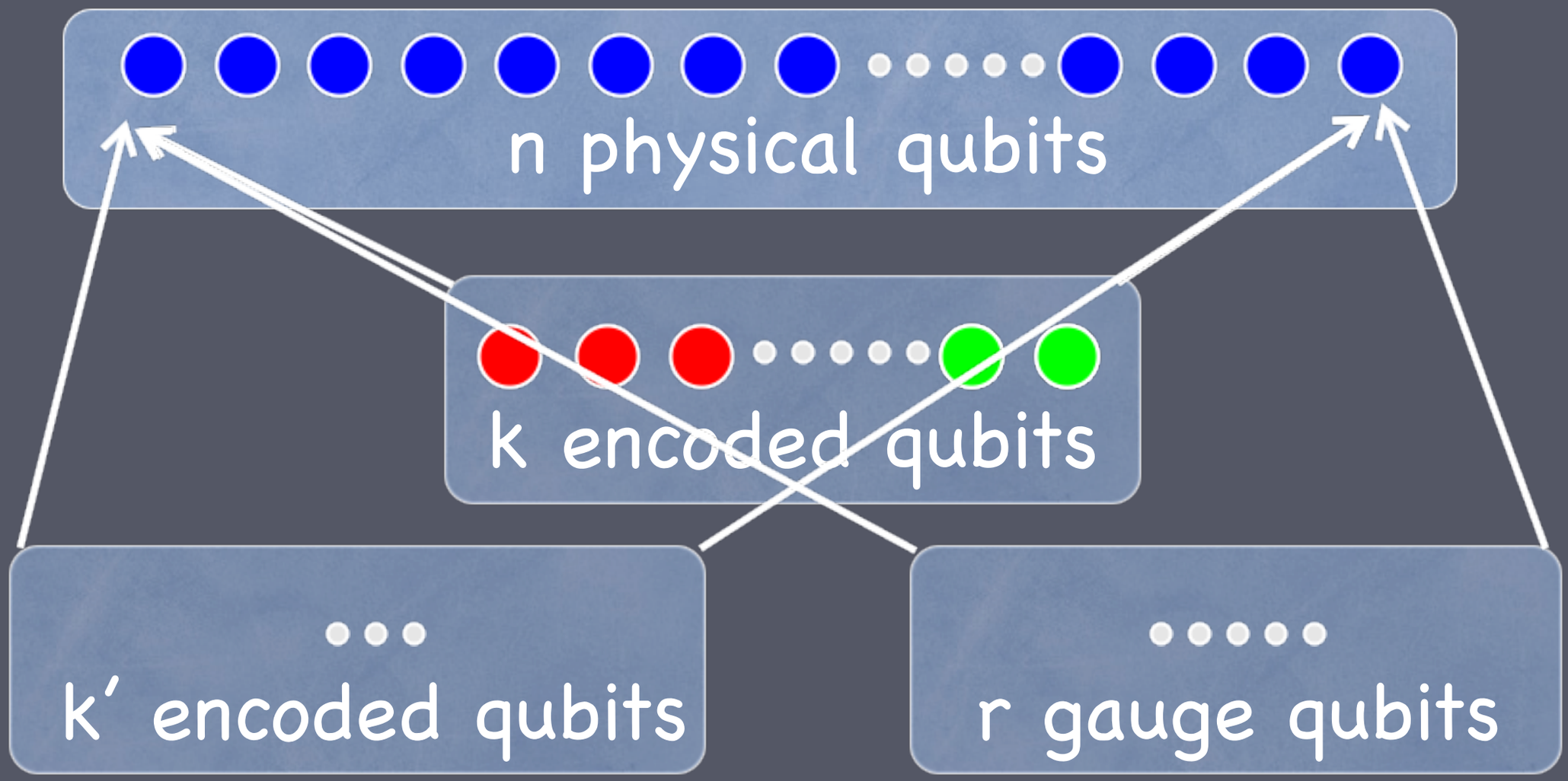
For every [[n,k,d]] stabilizer code, there exists a [[N,k,r,d]] stabilizer subsystem code, $N=O(n^2)$, whose syndrome measurements can be made using s-neighboring measurements (s constant).

To Understand:

1. stabilizer subsystem codes
2. measurement based quantum computing
3. spaced-time locality in QEC

1. Stabilizer Subsystem Code

[[n,k,d]] Stabilizer Subspace Code

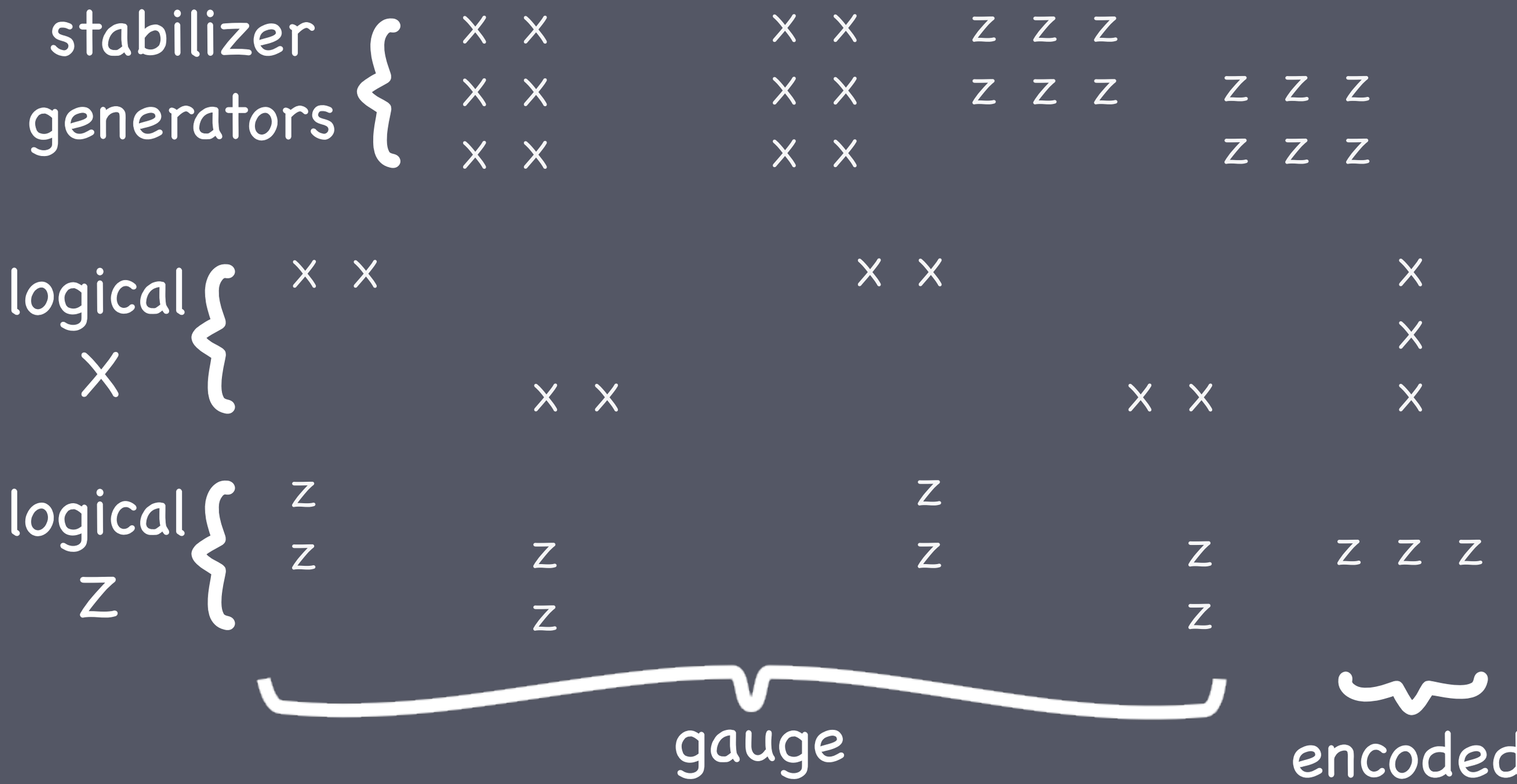


[n,k′,r,d] Stabilizer Subsystem Code

[David Poulin, arXiv:quant-ph/0508131]

A Subsystem Code

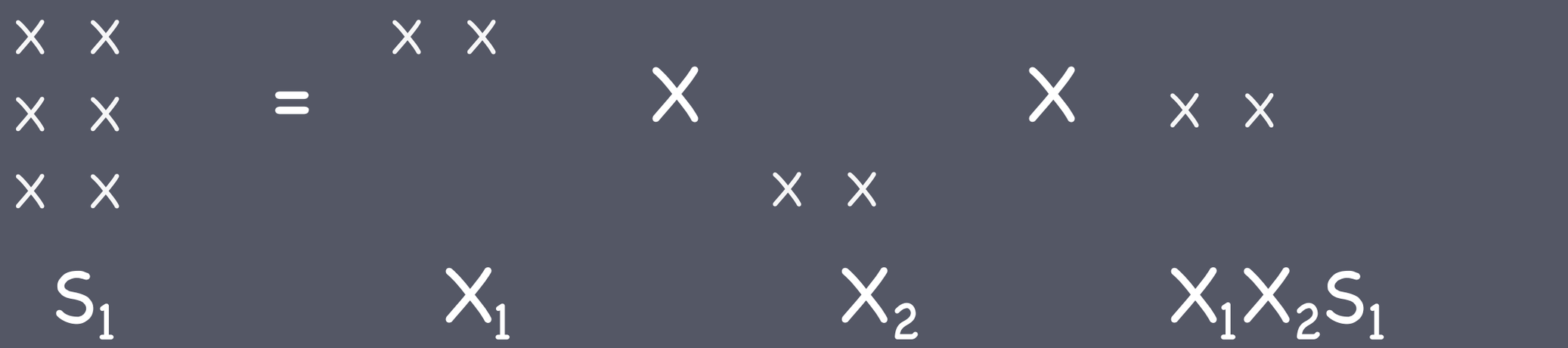
[Bacon, arXiv:quant-ph/0506023]



[[9,1,4,3]] stabilizer subsystem code

A Subsystem Code

stabilizer generators { S_1 S_2 S_3 S_4



We can measure S_1 by measuring 2-local operators that are products of gauge and stabilizer operators

6-local → 2-local

Stabilizer Moves

S_1	S_2	....	S_{n-k-r}	z_1	z_2	...	z_r	S_{n-k-r}	Z_1	Z_2	.... Z_k
				x_1	x_2	...	x_r	X_0	X_1	X_2	.... X_k

turn a stabilizer into a logical operator

S_1	S_2	....	S_{n-k-r}	z_1	z_2	...	z_r	Z_1	Z_2	....	Z_k
				x_1	x_2	...	x_r	X_1	X_2	....	X_k

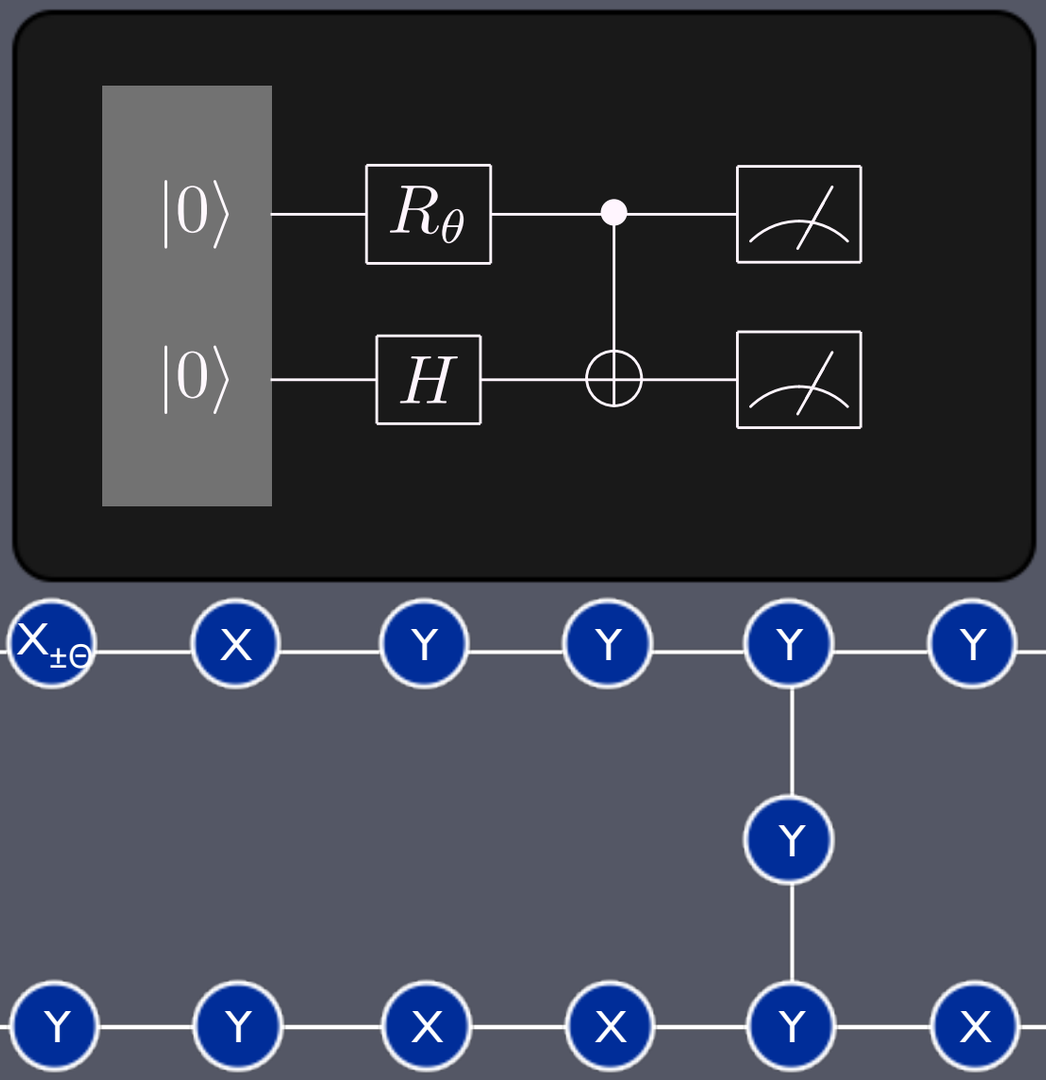
stabilizer gauge logical

turn a stabilizer into a gauge operator

S_1	S_2	....	$S_{n-k-r-1}$	S_{n-k-r}	z_1	z_2	...	z_r	Z_1	Z_2	.... Z_k
				x_0	x_1	x_2	...	x_r	X_1	X_2	.... X_k

2. Measurement Based QC

[Raussendorf and Briegel, Phys. Rev. Lett. 86, 5188 (2001)]



Adaptive measurements

Create entangled state

Entangling gates

Single qubit circuit

Initial state described locally + measurements are all local

MBQC Wire to Code



Graph state 1D line: common +1 eigenstate of graph state vertex operators

S_1=XZII….

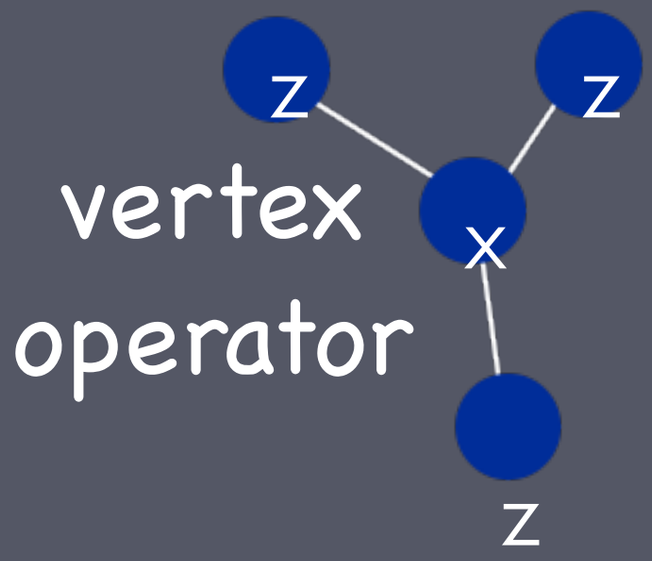
S_2=ZXZI….

S_3=IZXZ….

…

S_{n-1}=….IZXZ

S_n = …IIXZ



MBQC Wire to Code



Remove end stabilizer from stabilizer group and promote it to a logical operator

Stabilizer generators

~~S_1=XZII....~~

S_2=ZXZI....

S_3=IZXZ....

...

S_{n-1}=....IZXZ

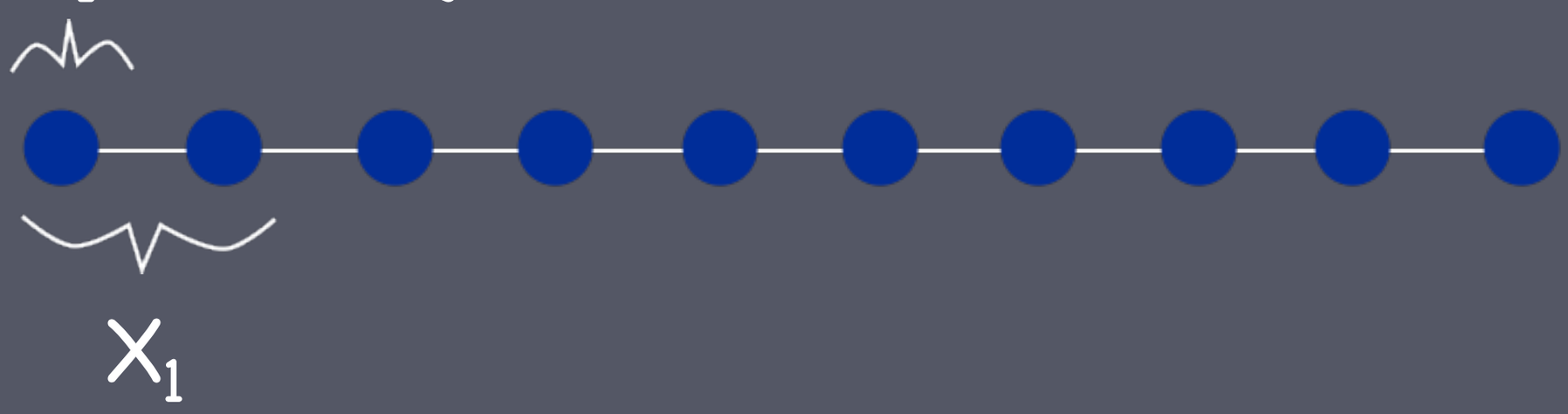
S_n=....IIXZ

Logical operators

X_1=XZIIII, Z_1=ZIIIII

One encoded distance 1 qubit

Qubit Where?



Stabilizer generators

~~S_1=XZII....~~

S_2=ZXZI....

S_3=IZXZ....

...

S_{n-1}=....IZXZ

S_{n-2}=....IIXZ

Logical operators

X_1=XZIIII..., Z_1=ZIIIII...

S_2Z_1=IXZIII...

S_4Z_1=IXIXZI...

modulo stabilizer information can be accessed across many different r-local measurements

MBQC Wire to Code



Make S_2 a gauge qubit

Stabilizer generators

~~S_1=XZII....~~

~~S_2=ZXZI....~~

S_3=IZXZ....

...

S_{n-1}=....IZXZ

S_n=....IIXZ

Logical operators

~~X_1=XZII..., Z_1=ZIII...~~

Gauge operators

x_1=XIII..., z_1=ZXZI...

MBQC Wire to Code



Make S_2 a gauge qubit

Stabilizer generators

~~S_1=XZII....~~

~~S_2=ZXZI....~~

S_3=IZXZ....

...

S_{n-1}=....IZXZ

S_n=....IIXZ

Logical operators

X_1=XZII..., Z_1=IXZI...

Gauge operators

x_1=XIII..., z_1=ZXZI...

X_1x_1=IZII...

Z_1 moved over a bit...

MBQC Wire to Code



Stabilizer generators

none

Logical operators

X_1=XIXI…XIX, Z_1=IXIX…IXZ

Modulo gauge operators localized to last qubit

Gauge operators

Generated by

XIII..II, IXII…II, … , IIII.,.XI

ZXZI..II, IZXZ…II, … , IIII…ZX

MBQC Wire to Code



Stabilizer generators

S_1=XIXI…XIX

Logical operators

none

Measure S_1 by series of single qubit X measurements (gauge)

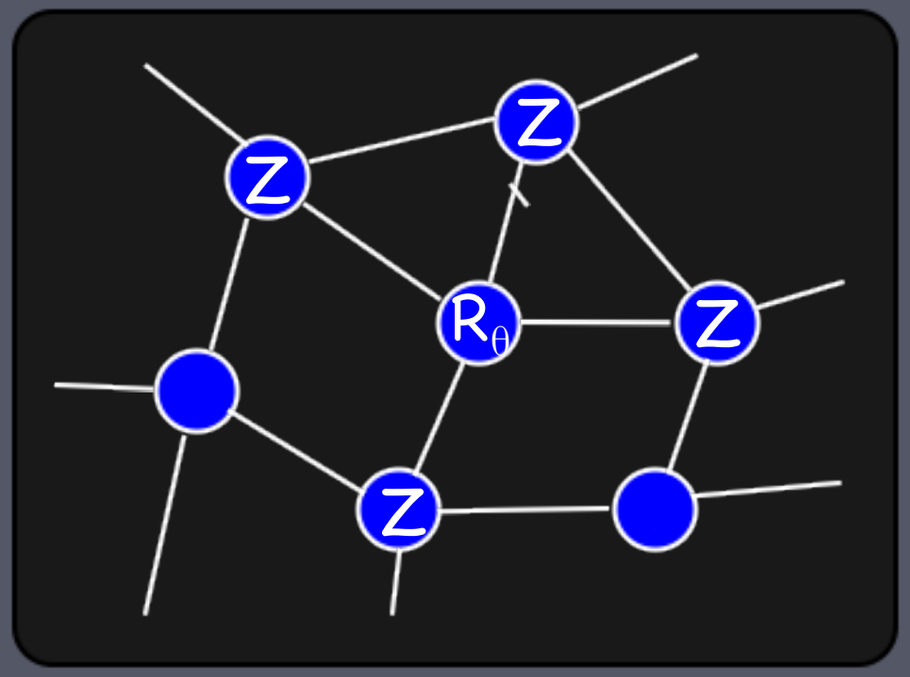
Gauge operators

Generated by

XIII..II, IXII…II, … , IIII.,.XI

ZXZI..II, IZXZ…II, … , IIII…ZX

Twisted Graph State



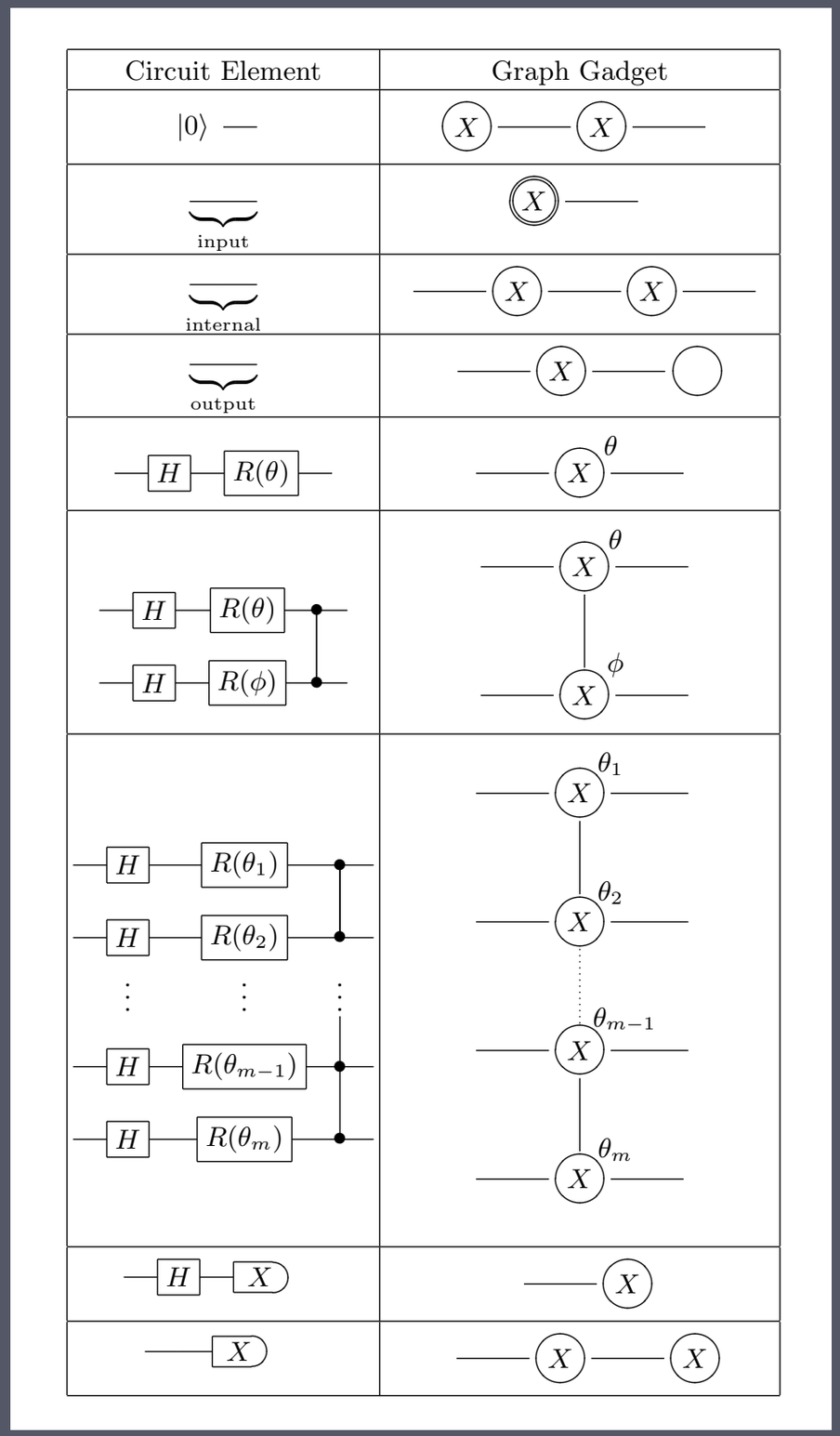
$$= \quad [R_\theta]_v \quad \prod_{w|(v,w)\in\mathcal{E}} \quad [Z]_w$$

$$R_\theta = \cos\theta X + \sin\theta Y$$

"Vertex operator with angle θ"

Twisted Graph State: is common +1 eigenstate of all vertex operators (one per vertex)

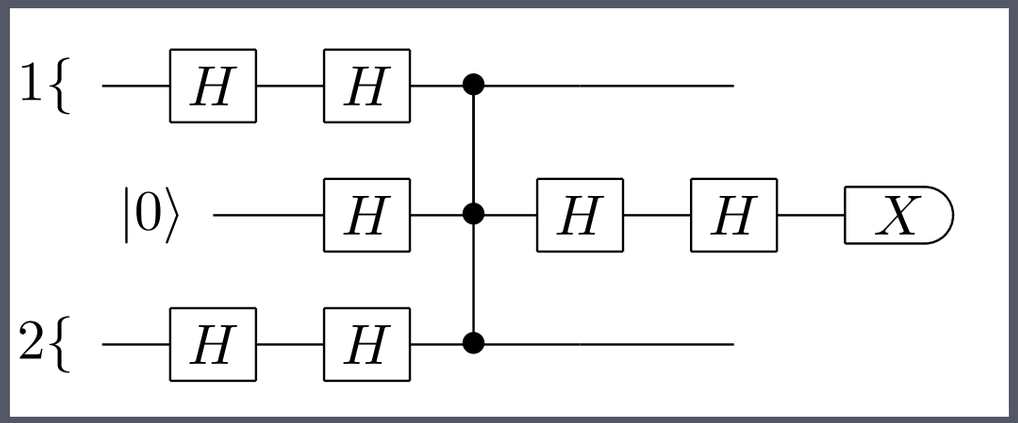
QC to Twist Graph State

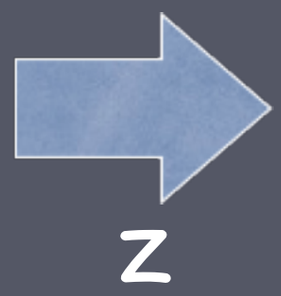


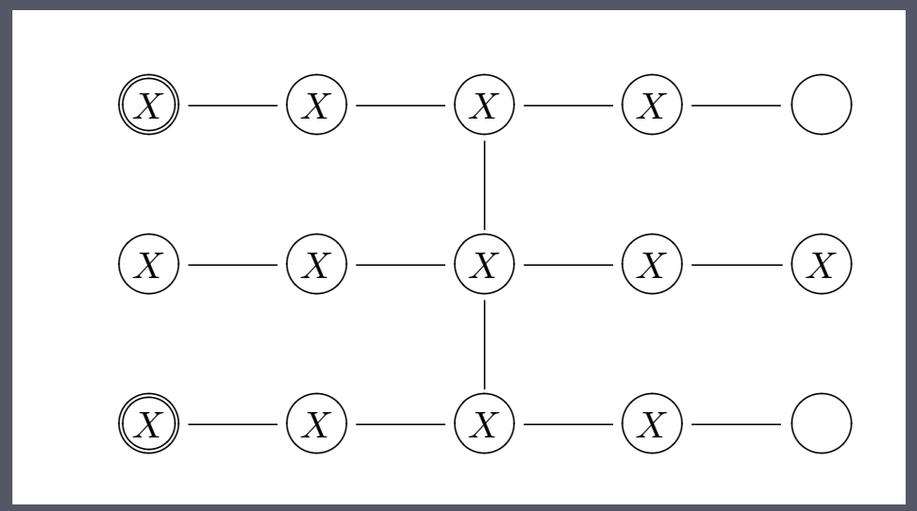
Quantum circuit made up of preparations, gates, and measurements

Graph describing vertex operators and places where X measurements would be made to enact MBQC for the quantum circuit

Encoded MBQC Example

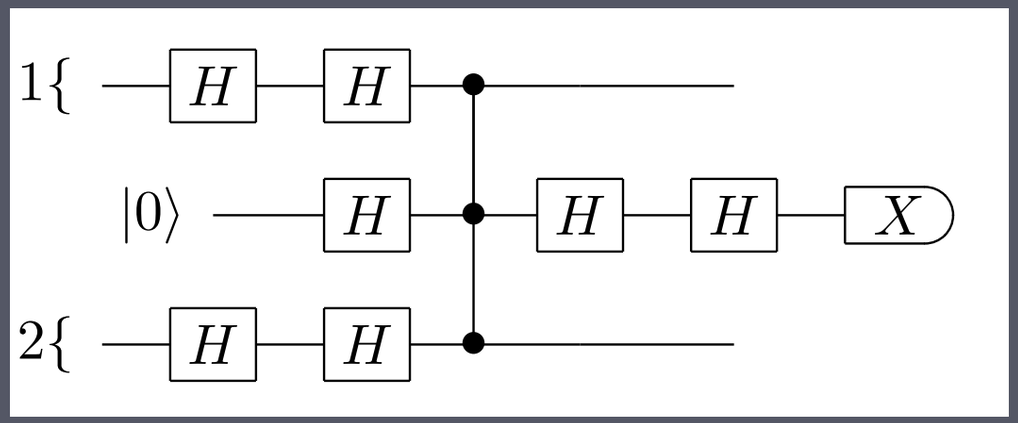


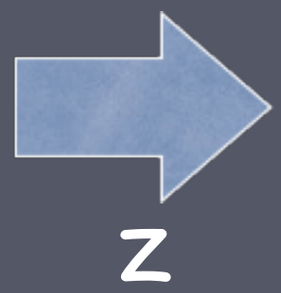


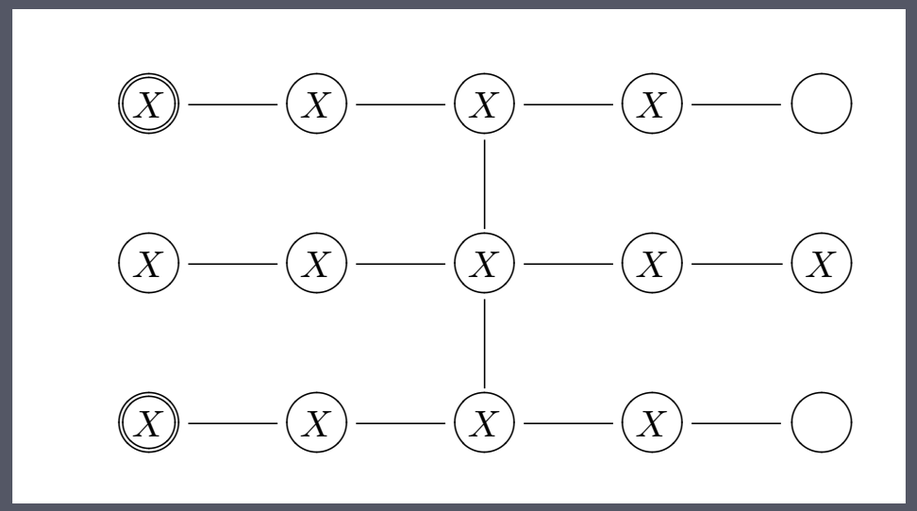


1. vertex operator for every non-double-circled node
2. starting state is +1 common eigenspace of these vertex operators
3. input information encoded @ double-circled nodes
4. X measurements occur on all vertices labeled X
5. Output measurements at X's, unmeasured output qubits at non-X-ed nodes

Encoded MBQC Example



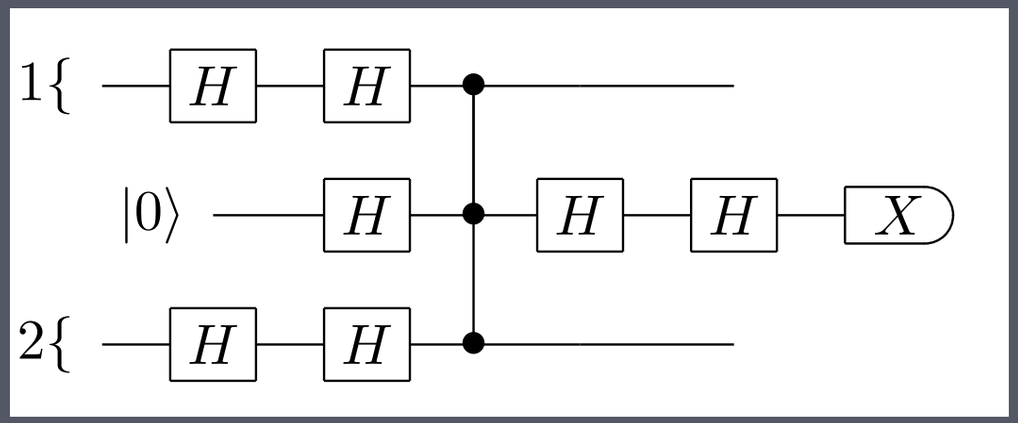


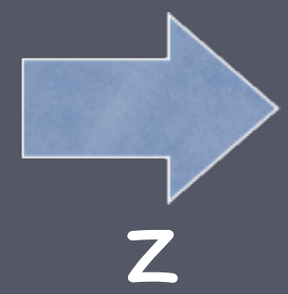


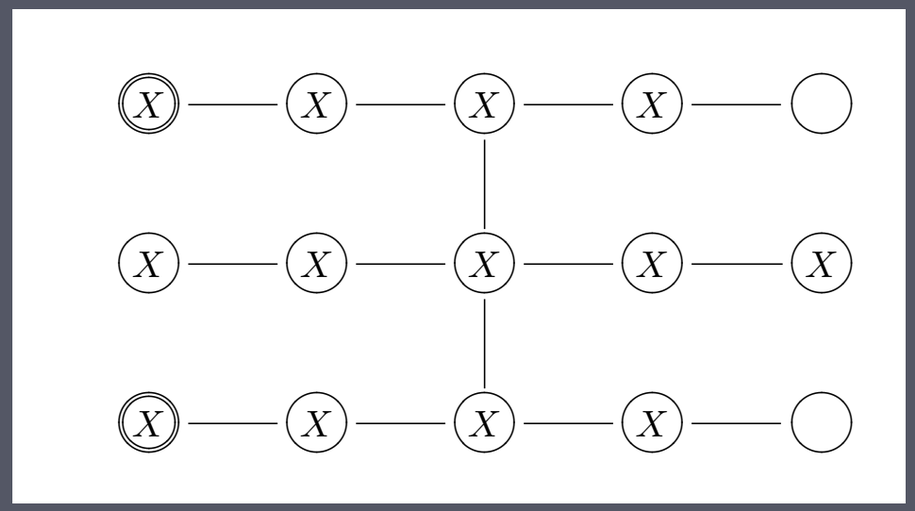
1. vertex operator for every non-double-circled node
4. X measurements occur on all vertices labeled X

Instead of performing MBQC we can instead consider the code generated by these two sets of operators.

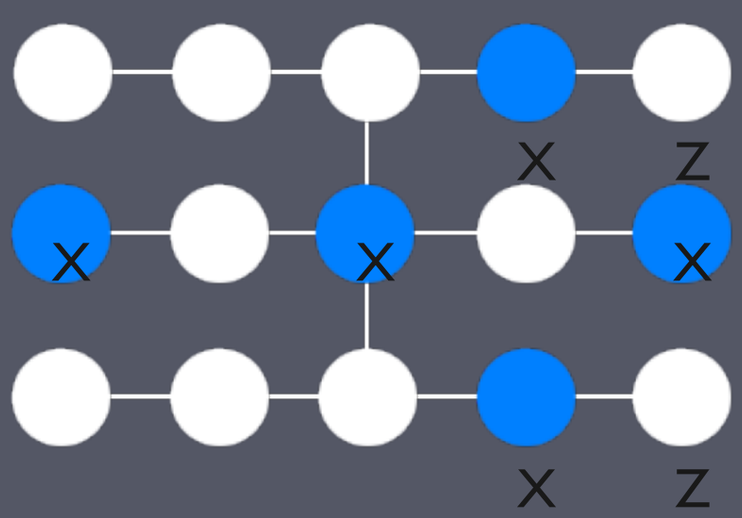
In Action



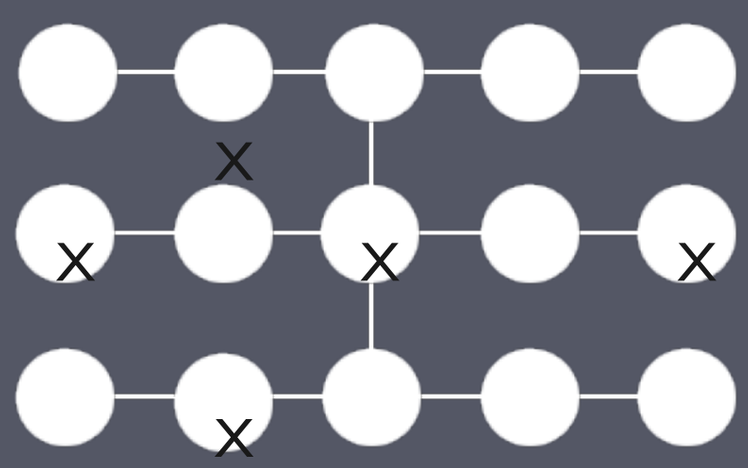




$S_1 =$



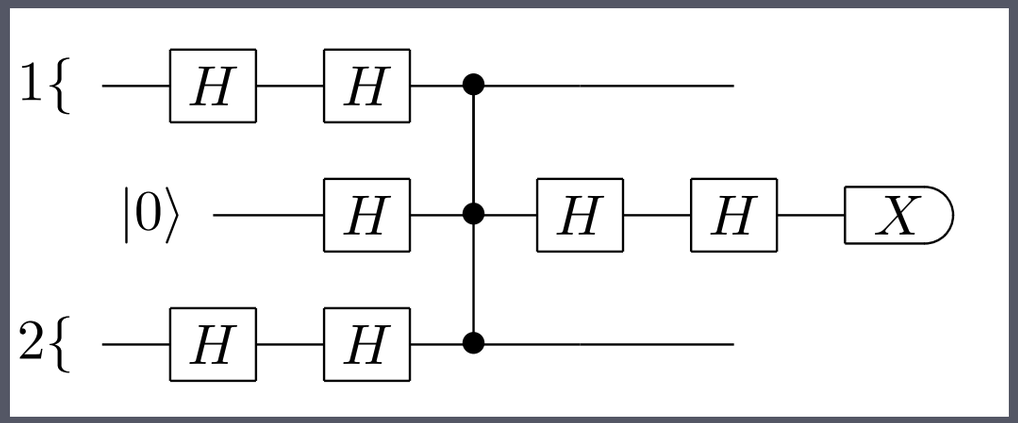
$S_2 =$

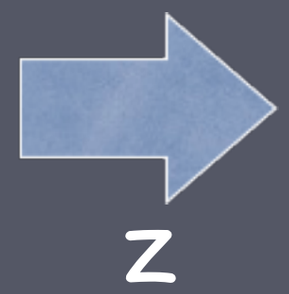


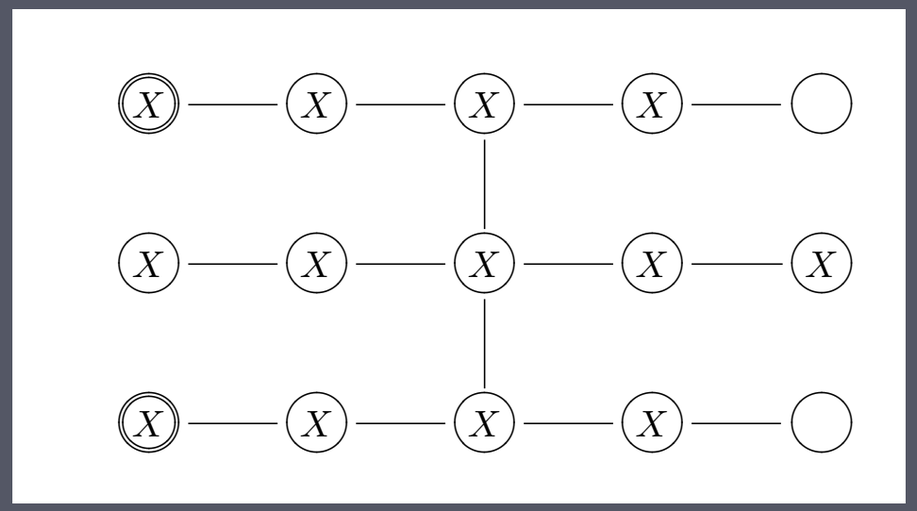
S_1 measured by measuring blue vertex operators.

S_2 represents that preparation and measurement coincide.

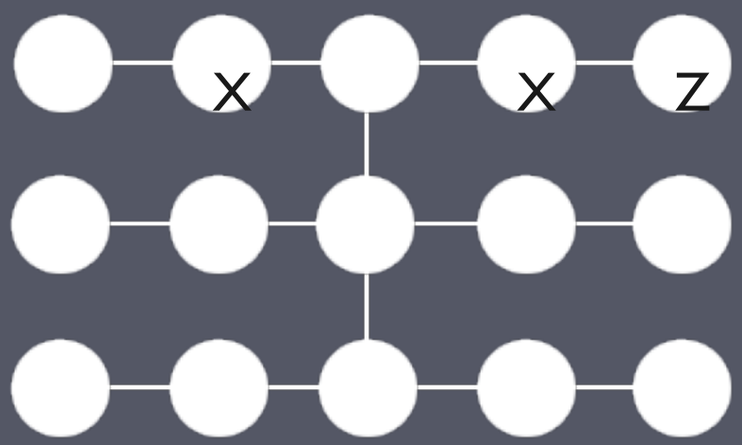
In Action



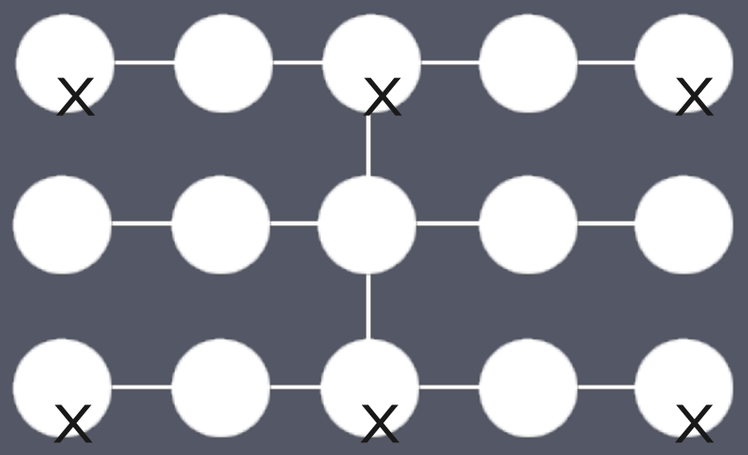




$\bar{Z} =$



$\bar{X} =$

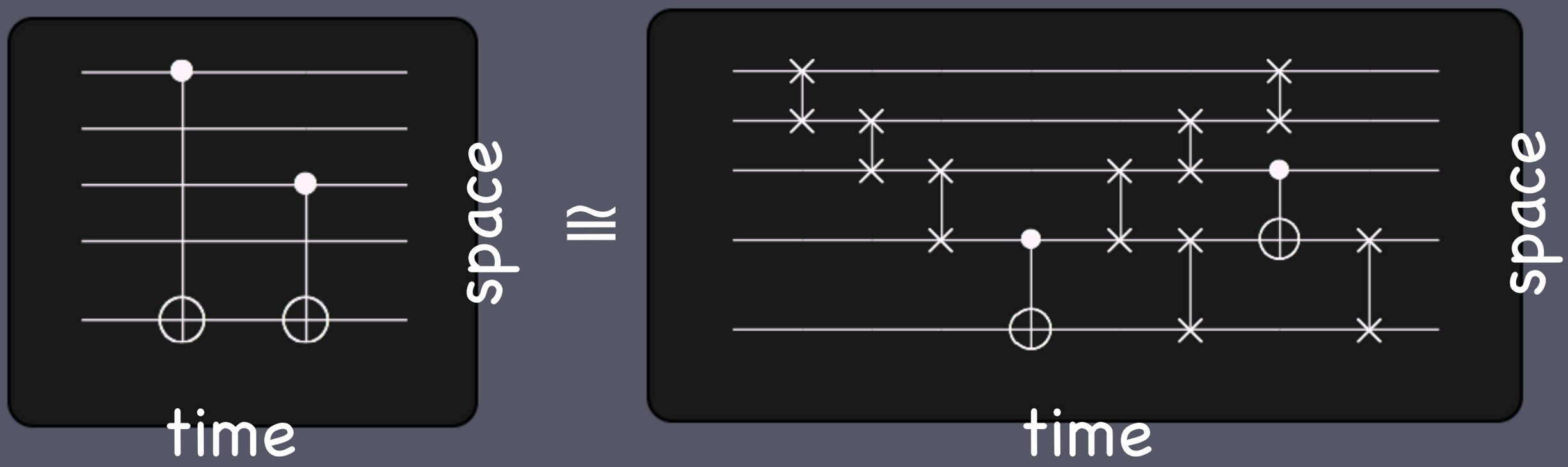


Modulo gauge and stabilizer group logical Z can be made weight 1, but logical X can only be made weight 2

3. Sidebar: Space-Time Neighboring

Any quantum circuit can be made to be neighboring in space-**time** for any spatial dimension D

Example in 1D



Space-Time QEC

Fault-tolerant quantum error correction can be done using s-neighboring locality in space-time while maintaining a threshold.

(1D requires next-nearest neighbor gates)

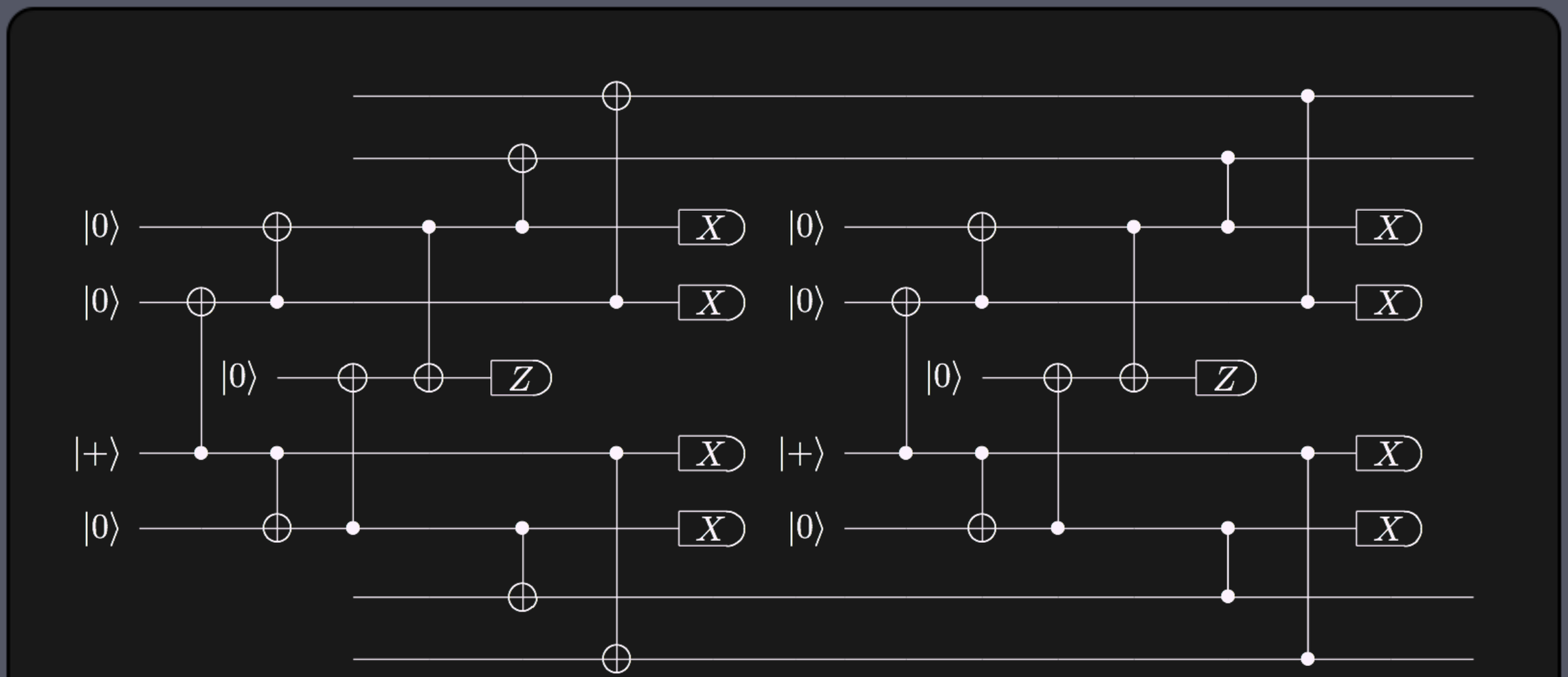
[Aharonov, Ben-Or, arXiv:quant-ph/9906129]
[Gottesman, arXiv:quant-ph/9903099]



THE CONSTRUCTION

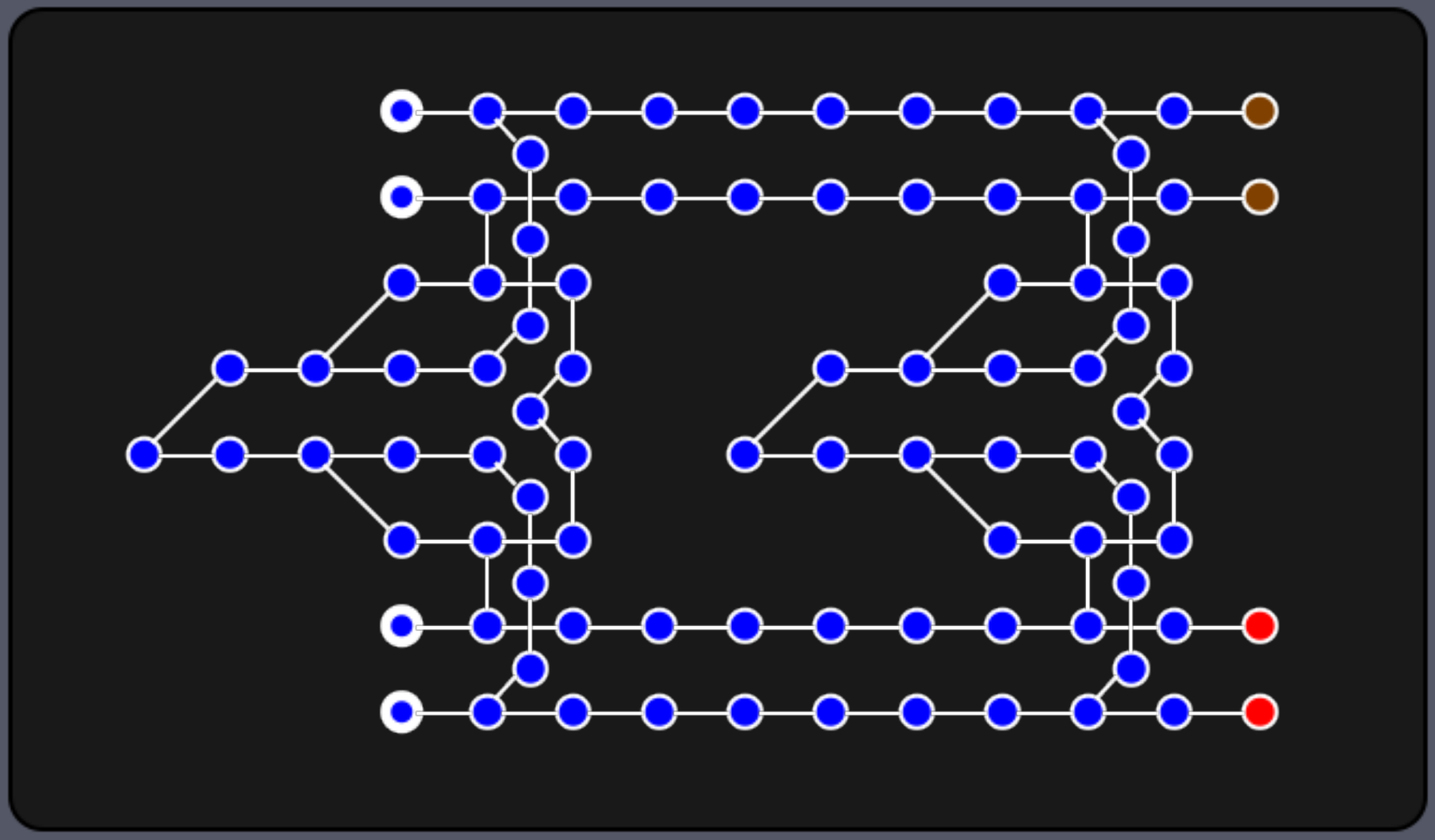
Input: [[n,k,d]] stabilizer code

1. Construct fault-tolerant syndrome measuring quantum circuit that is s-neighboring in space-time on lattice in dimension D (D>1)



CONSTRUCTION

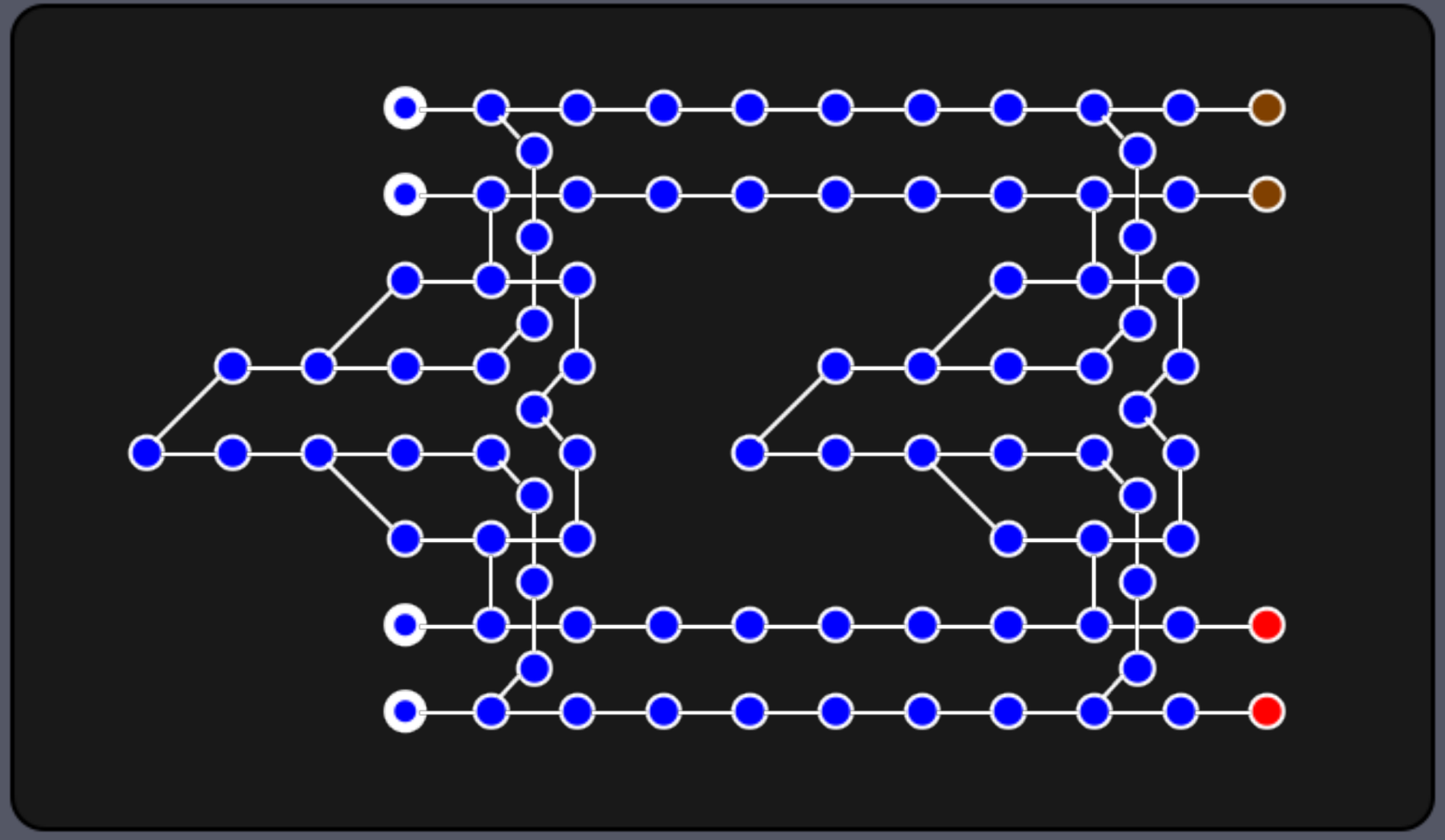
2. Convert this circuit to the scheme that would be used to implement it in MBQC



[Related: Rausendorf, Harrington, Goyal 2006]

CONSTRUCTION

3. Take operators from MBQC construction (vertex operators and X operators for measurements) and identify (a) stabilizer subgroup of these operators and (b) gauge qubits

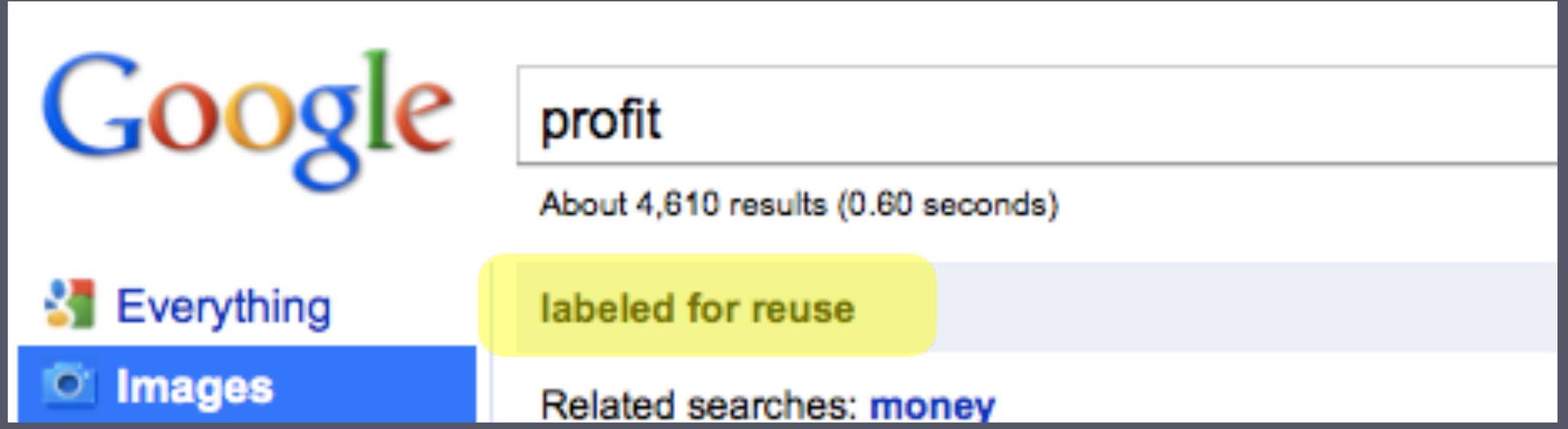


$X^{\otimes 4}$ and $Z^{\otimes 4}$ stabilizer localized on red qubits

Profit

For every [[n,k,d]] stabilizer code, there exists a [[N,k,r,d]] stabilizer subsystem code, $N=O(n^2)$, whose syndrome measurements can be made using s-neighboring measurements (s constant).

$$$$



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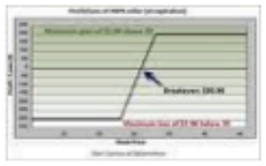
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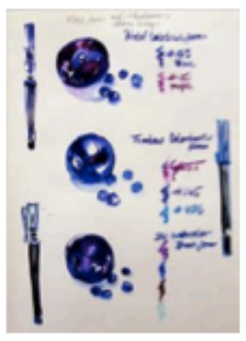
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FAQ

Q: Why fault-tolerant syndrome measuring circuits?
- These maintain the distance of the code
Q: But aren't fault-tolerant circuits probabilistic?
- Often yes. Use the circuit as if the measurement result turned out in your favor
Q: Do these codes have thresholds?
- Unknown, but likely given their relation to fault-tolerant circuits
Q: s-local for what s?
- Can be made 3-local

Consequences



arXiv:soon (after we launch)

Q-Dub Group Still Lives!

<u>Grads</u>

Isaac Crosson (Physics)
Lukas Svec (Physics)
Jijiang Yan (Physics)
Kamil Michnicki (Physics)
David Rosenbaum (CS)
Paul Pham (CS)
Kevin Zatloukal (CS)

<u>Faculty</u>

Aram Harrow (Pseudo-prof)
Steve Flammia (Scientist!)

<u>Undergrads</u>

Zakk Webb (Physics/CS/Math)
Kate Liotta (Physics/CS)
Jonathan Shi (Physics/CS)
Melanie Jensenworth (Math/CS)
Jeffrey Booth (CS)
Rob McClure (CS)
Harshad Petwe (CS)

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